Sparse-Prior Semantic Segmentation for Digital Pathology:

Adapting a CT Bone-Segmentation Framework to H&E Nuclei Segmentation on PanNuke

Dr. K. Mithra

Independent Researcher · Scottsdale, Arizona

September 2026

Abstract. Bonnet is a recently proposed pipeline for CT bone segmentation that restricts computation to a sparse set of voxels selected by a Hounsfield-unit intensity prior, processed with a sparse convolutional U-Net and reassembled with overlapping, Gaussian-weighted sliding windows. This report investigates whether the same design transfers to a structurally different domain: nuclei segmentation in H&E-stained digital pathology. A stain-deconvolution prior (hematoxylin concentration, and separately total optical density) is substituted for the CT intensity window, calibrated on training data to retain 98% of nuclear pixels, and paired with a matching sparse U-Net. The pipeline is trained and evaluated end to end on the PanNuke dataset (fold 1 train / fold 2 validation / fold 3 test) against a dense U-Net baseline of identical width, normalization, activation, and loss. The two masking priors sacrifice 31–36% of pixels (a third, unmasked sparse control retains 100% as an ablation), yet the sparse pipeline is comparable to or slightly below the dense baseline on nuclei-macro Dice in every configuration, while running 5–6× slower per image. An external, no-fine-tuning cross-dataset evaluation on CPM-17 (32 held-out histology tiles, substituted for the originally intended CoNSeP set after that dataset's public download was placed behind an institutional sign-on) reproduces the same ordering: the dense baseline is again both the most accurate (83.6% foreground Dice) and by far the fastest configuration. The result is reported as a negative finding, consistent across both the internal and external test sets: unlike CT bone, which occupies a genuinely small fraction of a 3D volume, stained nuclei occupy the majority of a tissue patch, so a sparsity prior removes little real computation while adding sparse-tensor bookkeeping overhead. The report documents the calibration method, architecture, training protocol, and full metrics, and outlines the conditions (whole-slide-level tissue/background separation rather than patch-level nuclei/stroma separation) under which the same idea would be expected to pay off.

1. Introduction

Whole-slide images (WSIs) in digital pathology are gigapixel-scale, and the diagnostically relevant structures — nuclei, glands, mitotic figures — occupy only a fraction of any given field. This has motivated interest in segmentation pipelines that concentrate computation on informative regions rather than processing every pixel uniformly, mirroring a design pattern already explored in 3D medical imaging. Recent work on CT bone segmentation (referred to here as *Bonnet*) restricts a sparse convolutional network to voxels passing a fixed Hounsfield-unit (HU) window, reconstructs the full volume with overlapping sliding windows fused by a Gaussian kernel, and reports large speed gains with no loss of accuracy when generalizing across CT datasets without fine-tuning.

This report asks a direct question: does the same architectural pattern transfer when the intensity prior is replaced with a domain-appropriate stain prior, and the 3D sparse U-Net is replaced with its 2D analogue? Digital pathology is a natural test case, since H&E staining gives a principled, unsupervised way to estimate a "nuclear prior" per pixel via stain deconvolution, playing the same structural role that the HU window plays for bone. The contribution here is not a new state-of-the-art segmentation network; it is a careful, like-for-like methodological transfer study, with an honestly reported outcome, run under a controlled architecture-matched comparison.

2. Related Work

Sparse priors for efficient segmentation. The Bonnet pipeline for CT bone segmentation thresholds Hounsfield units to obtain a sparse point cloud, applies a sparse 3D U-Net (submanifold and strided sparse convolutions, sparse instance normalization), and reassembles predictions from overlapping windows using Gaussian-weighted fusion. It reports strong cross-dataset generalization without fine-tuning against dense nnU-Net baselines.

PanNuke. PanNuke is a pan-cancer histology dataset of 256×256 H&E patches spanning 19 tissue types, with per-pixel instance masks for five clinically relevant nuclear categories (neoplastic, inflammatory, connective, dead, epithelial), released as three pre-defined folds for cross-validation.

CoNSeP and CPM-17. CoNSeP is a colorectal-adenocarcinoma nuclear segmentation and classification dataset commonly used alongside PanNuke to assess cross-dataset generalization of nucleus segmentation models; it uses a different nuclear class taxonomy, so cross-dataset comparisons are typically reduced to a binary nucleus-versus-background task. CoNSeP was the originally intended external test set for this study; at the time of evaluation its public download had been placed behind an institutional sign-on wall with no available mirror, so the external evaluation reported here instead uses CPM-17, a nuclear instance-segmentation dataset of H&E tiles released alongside the same HoVer-Net line of work and distributed in an equivalent image-plus-instance-mask format. Like CoNSeP, CPM-17 carries no class taxonomy compatible with PanNuke's five nuclear categories, so it is evaluated identically: as a binary nucleus-versus-background task.

Stain deconvolution. The hematoxylin and eosin optical-density unmixing method of Ruifrok and Johnston provides a standard way to recover a per-pixel hematoxylin concentration channel from an RGB H&E image, which is used here as the stain-based nuclear prior, analogous in role to the CT HU window.

3. Method

3.1 Sparse prior construction

Two stain-derived priors were evaluated, plus a no-prior control:

- **Hematoxylin prior (`hema`):** the RGB image is converted to optical density and unmixed against reference hematoxylin/eosin/residual stain vectors; the hematoxylin channel is used as the per-pixel nuclear score.
- **Tissue prior (`tissue`):** the mean optical density across all three channels is used as a coarser, stain-agnostic tissue-density score.

- **No prior (`raw`):** every pixel is retained (100% density), used as an ablation to isolate the effect of the sparse network architecture itself from the effect of masking.

For each prior, a threshold τ is calibrated once on the training fold by sampling 400 training images, computing the prior score at every ground-truth nuclear pixel, and setting τ to the score quantile that retains 98% of nuclear pixels (a fixed target recall, chosen analogously to the way the CT HU window is chosen to contain essentially all bone voxels). Per-channel RGB normalization statistics and prior-score normalization statistics are computed jointly over the retained pixels at this threshold.

3.2 Sparse windowing and Gaussian fusion

At inference, the calibrated mask is applied to the full image, and the retained pixels are processed through overlapping windows of size 128×128 with 50% stride overlap. Each window's softmax output is weighted by an isotropic Gaussian centered on the window ($\sigma = 0.5\times$window size) before being accumulated into a shared probability buffer, and the final label at each pixel is the argmax of the fused, weight-normalized distribution:

$$a_t(x) = \exp\left(-\|x - m_t\|^2 / (2(\sigma w)^2) \right) \quad \bar{P}(x) = \Sigma_t\, a_t(x) \cdot P_t(x) \,/\, \Sigma_t\, a_t(x)$$

Pixels never covered by any retained window (i.e., outside the prior mask everywhere) are assigned to background by construction.

3.3 Network architectures

Two architectures were trained under matched conditions:

- **Sparse U-Net:** a 4-level 2D sparse U-Net (widths 32/64/128/256) built from submanifold and strided sparse convolutions with sparse instance normalization, LeakyReLU(0.01) activations, and a 2-layer MLP classification head, operating only on the retained sparse pixels.
- **Dense U-Net (baseline):** a conventional dense 2D U-Net of identical depth, channel widths, instance normalization, and activation, operating on every pixel. This is the direct analogue of the dense nnU-Net baseline in the source CT study.

Both networks total approximately 2.3–2.5 million parameters. Both are trained with the same combined loss:

$$L = \mathrm{CE}_{\text{label-smoothed (0.1)}}(\hat{y}, y) + (1 - \mathrm{SoftDice}(\hat{y}, y))$$

Soft Dice is computed over the five foreground nuclear classes only, averaged, so that the very small "dead cell" class does not vanish from the objective purely due to class imbalance in the cross-entropy term.

4. Experimental Setup

Dataset	PanNuke — Fold 1 (train), Fold 2 (validation), Fold 3 (held-out test)
Task	6-way semantic segmentation (background + 5 nuclear classes)
Training budget	40 epochs × 150 mini-batches/epoch, batch size 8, 128×128 crops

Crop sampling	33% foreground-biased (centered on a random nuclear pixel), 67% uniform random, with flips, 90° rotations, and mild color jitter
Optimizer	AdamW, weight decay 1e-4, One-Cycle learning-rate schedule, peak LR 2e-3
Model selection	Checkpoint with the highest validation nuclei-macro Dice across training
Hardware	Single NVIDIA T4 GPU (Google Colab)
Runs compared	`sparse+hema`, `sparse+tissue`, `sparse+raw`, `dense+raw`
External test set	CPM-17 test split, 32 H&E tiles, evaluated with no fine-tuning using the checkpoints trained on PanNuke fold 1

4.1 Representative training samples

Figure 1 shows four representative training patches (PanNuke fold 1, sample indices 0, 10, 50, and 100), each with its input H&E image alongside the corresponding ground-truth nuclear mask, collapsed here to a binary nucleus-versus-background view for clarity. These are raw data-and-annotation pairs, not model output, and are shown only to illustrate the variety the model is trained on: tissue composition, staining intensity, and nucleus size, shape, and density all vary considerably across patches, from the large, well-separated nuclei in the first and fourth samples to the sparser, more elongated (likely connective or spindle-cell) nuclei in the second and third.

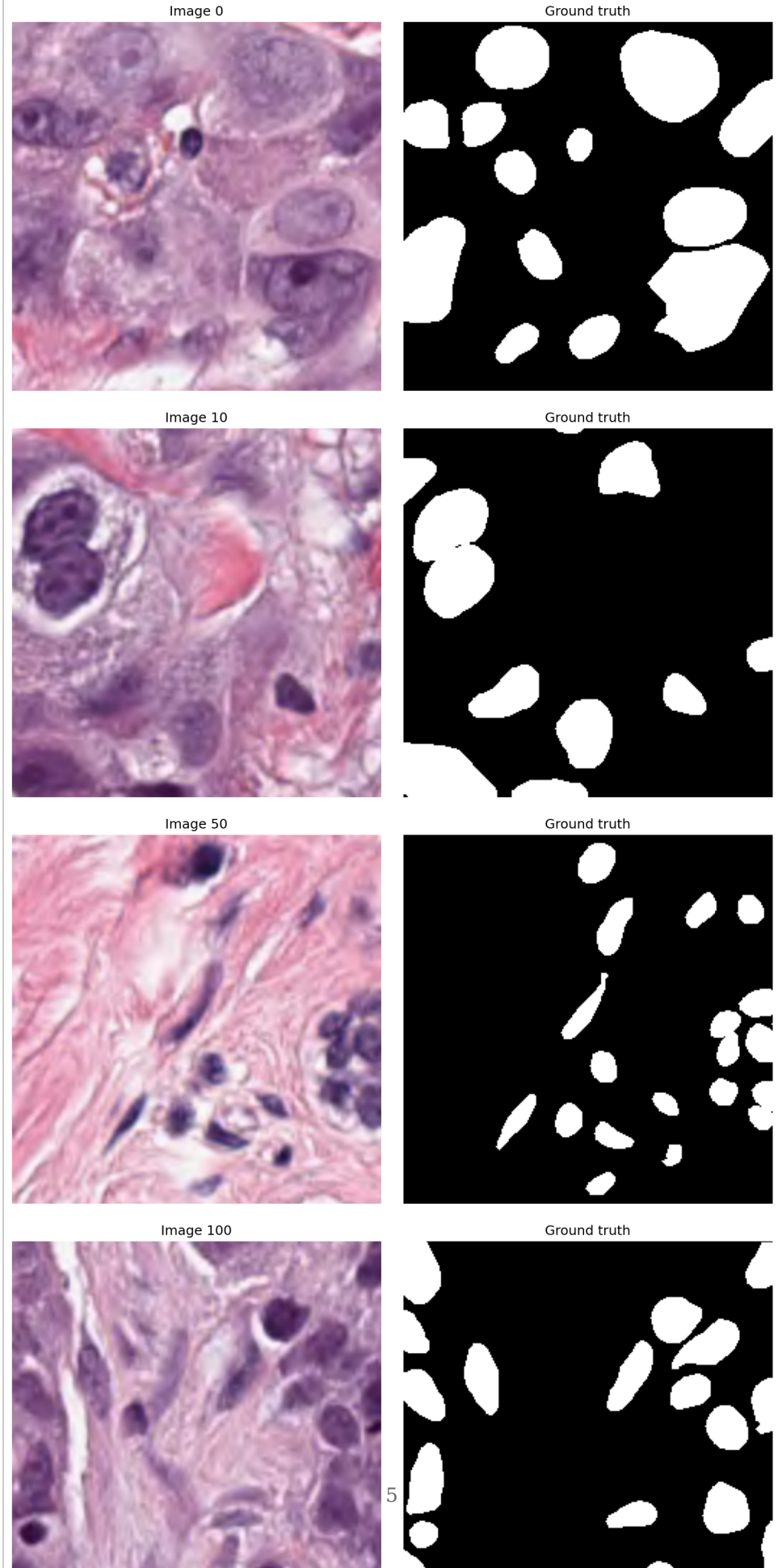

Image 0
Ground truth
Image 10
Ground truth
Image 50
Ground truth
Image 100
Ground truth

Figure 1. Four representative PanNuke fold-1 training samples (indices 0, 10, 50, 100), each shown as input H&E patch (left) and binary ground-truth nuclear mask (right). Illustrates data and annotation variety rather than model performance.

5. Results

Table 1 reports the internal test-set metrics (PanNuke fold 3, 2,722 images) for all four configurations: the fraction of pixels retained by the prior, the fraction of ground-truth nuclear pixels the prior mask actually covers (an upper bound on achievable Dice), pooled foreground Dice, macro-averaged Dice over the five nuclear classes, per-class Dice, mean forward-pass time per image, and the corresponding external (CPM-17) foreground Dice, evaluated with no fine-tuning using the same PanNuke-trained checkpoints.

Configuration	Prior	Kept px (%)	GT recall (%)	Foreground Dice	Nuclei macro Dice	Neopl.	Inflam.	Connect.	Dead	Epith.	CPM-17 fg Dice	Fwd. time (ms/img)
Dense U-Net	**none**	**100.0**	**100.0**	**0.797**	**0.385**	**0.648**	**0.458**	**0.414**	**0.000**	**0.408**	**0.836**	**12.2**
Sparse U-Net	raw (no mask)	100.0	100.0	0.776	0.380	0.655	0.458	0.411	0.000	0.375	0.813	75.6
Sparse U-Net	tissue (OD)	69.0	97.8	0.763	0.369	0.637	0.404	0.405	0.000	0.400	0.801	62.1
Sparse U-Net	hematoxylin	64.4	97.7	0.781	0.355	0.645	0.395	0.408	0.000	0.327	0.823	59.6

Table 1. PanNuke fold 3 (held-out) test results, plus external CPM-17 foreground Dice (Section 5.3). Dice values are class-pooled unless labeled per-class. "CPM-17 fg Dice" is binary (nucleus-vs-background), since CPM-17 carries no class taxonomy comparable to PanNuke's five nuclear categories. Forward time is measured on PanNuke patches and excludes the one-time prior-mask computation, which is under 8 ms/image in every configuration; CPM-17 forward times, measured separately on its larger tiles, are reported in Section 5.3.

> **Principal finding.** The dense U-Net baseline is the most accurate configuration (nuclei-macro Dice 0.385, ahead of every sparse variant, though only by 0.005 over the unmasked sparse control — within this study's run-to-run noise — and by a clearer 0.030 over the stricter hematoxylin-masked variant) and is the fastest by a wide, unambiguous margin (12.2 ms/image vs. 60–76 ms/image for the sparse variants, a 5–6× slowdown) on the internal PanNuke test set. Adding a stricter stain prior (hematoxylin, 64% retained) improved neither speed nor accuracy relative to the unmasked sparse network. This is the opposite ordering from the one reported for CT bone segmentation with an HU prior — and, as Section 5.3 shows, the same ordering reproduces on an entirely held-out external dataset with no fine-tuning.

5.1 Training behavior

All four configurations converged smoothly: training loss decreased monotonically over 40 epochs in every run, and validation nuclei-macro Dice rose from approximately 0.07–0.08 at epoch 1 to a plateau in the 0.35–0.42 range after roughly 20–30 epochs, with epoch-to-epoch fluctuation of about ±0.02 consistent with the

small per-epoch validation sample. No configuration showed divergence, instability, or overfitting collapse within the training budget used.

No model learned to segment the "dead cell" class (Dice 0.000 in every configuration). This class is a small minority of annotated nuclei in PanNuke, and under a fixed 40-epoch budget with foreground-biased but not class-balanced crop sampling, the combined loss did not surface enough dead-cell examples to learn a usable decision boundary for that class specifically; the other four nuclear classes were all learned with non-trivial Dice.

5.2 Qualitative results

Figure 2 shows four representative PanNuke fold-3 test patches for two configurations: the dense baseline (no prior, 100% of pixels retained) and the sparse hematoxylin-prior model (64% of pixels retained). Each panel shows, left to right: the input patch, the ground-truth nuclear mask, the model's prediction, and (for the sparse model) the actual masked input the network receives, with unmasked-out pixels blacked out.

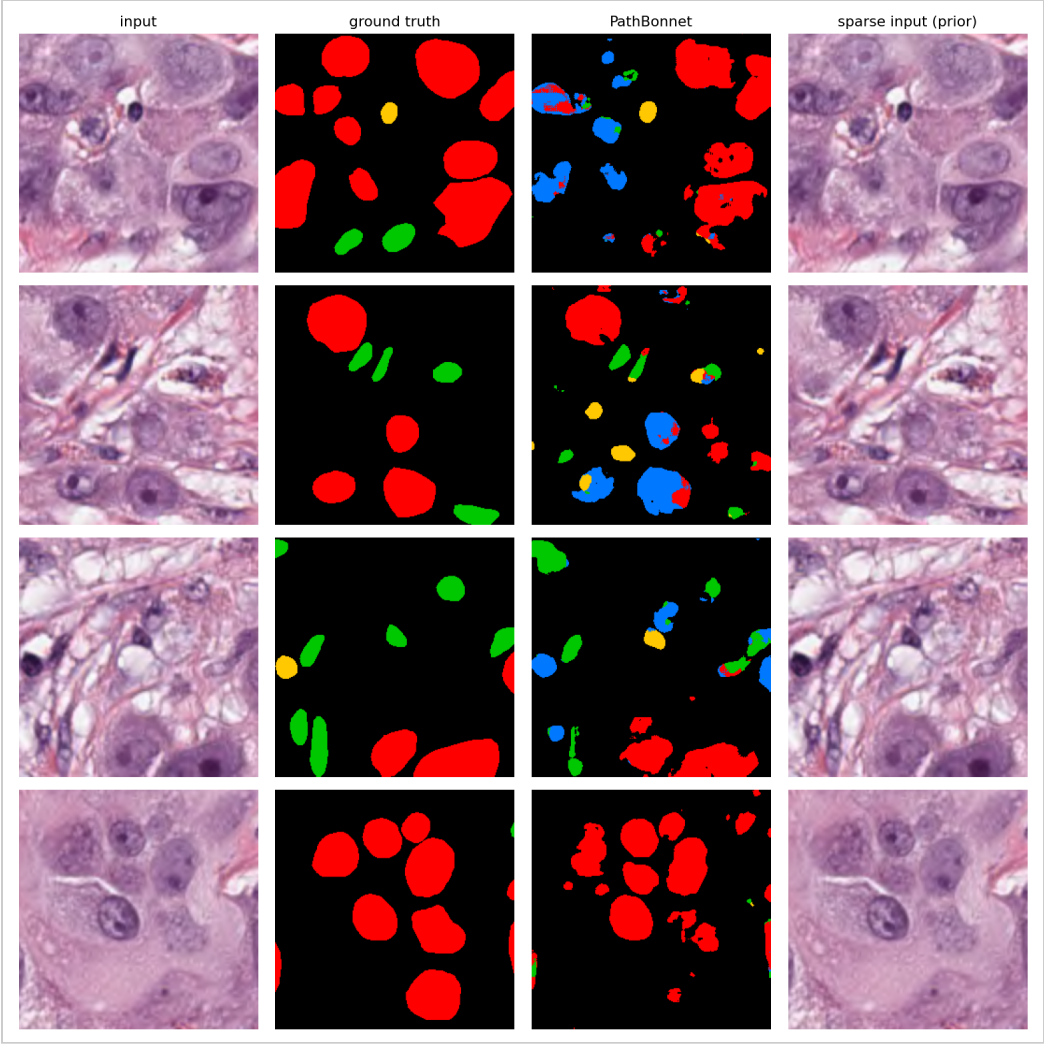


Figure 2a. Dense U-Net (no prior, 100% pixels retained). Columns: input / ground truth / prediction / sparse-mask input (identical to input, since no masking is applied). The third column's "PathBonnet" header, printed automatically by the evaluation script, labels the model's output for whichever configuration is shown — here, the dense U-Net.

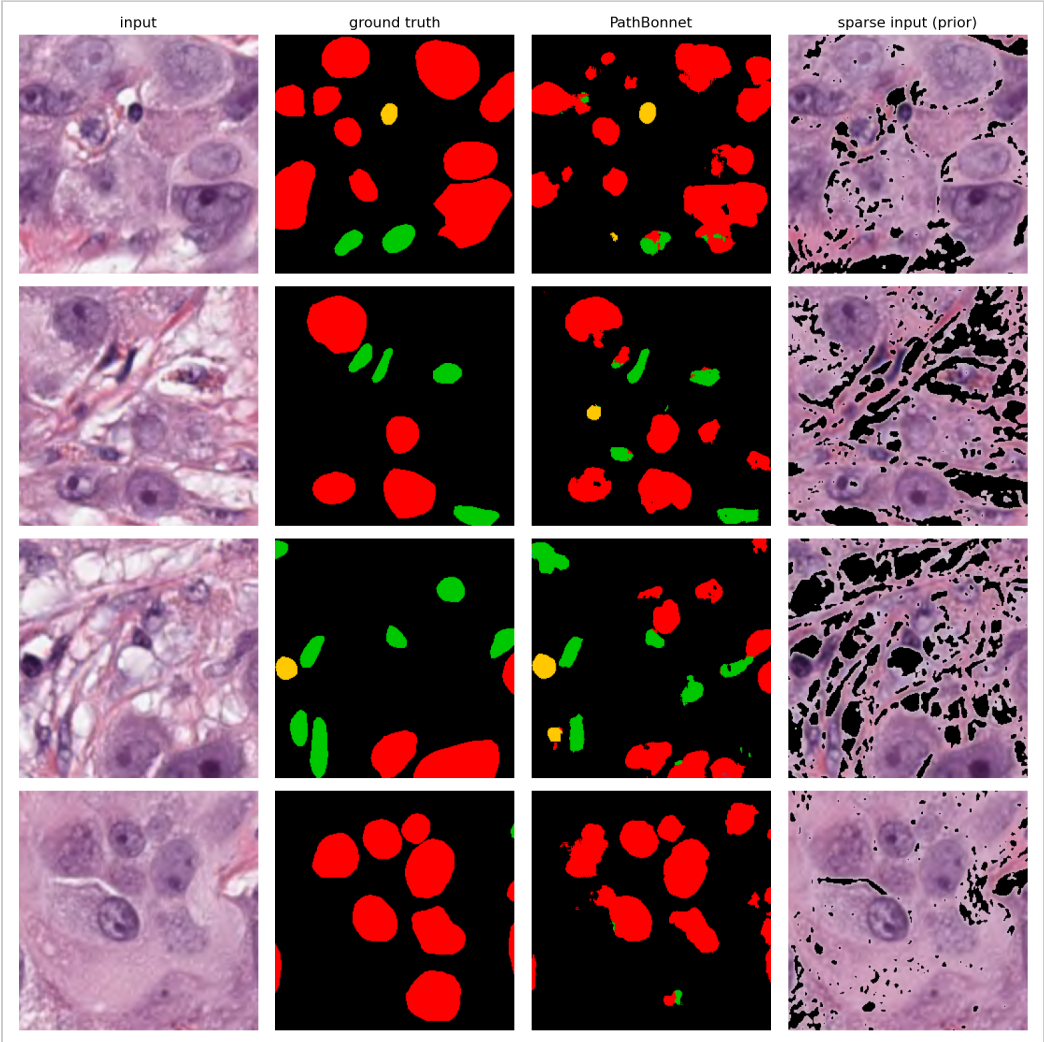


Figure 2b. Sparse U-Net with hematoxylin prior (64% pixels retained). The fourth column shows the actual masked input the network receives; black regions are excluded from computation. Predominantly eosin-stained stromal regions are correctly excluded while nuclei are retained, consistent with the 97.7% ground-truth mask recall reported in Table 1.

Two qualitative patterns are visible across both configurations, consistent with the quantitative results in Table 1. First, large, well-separated nuclei (typically neoplastic, shown in red) are segmented with good boundary agreement in both models. Second, both models produce spurious epithelial-class (blue) predictions in regions with no corresponding ground-truth annotation, most visible in the top-left and second-row patches of both panels; since this appears in the dense baseline as well as every sparse variant, it reflects a shared limitation of the training budget and class balance rather than an effect specific to sparsification, and is consistent with the moderate epithelial per-class Dice (0.33–0.41) reported in Table 1.

5.3 External generalization (CPM-17, no fine-tuning)

To test whether the internal result reflects something specific to PanNuke rather than the sparse-versus-dense comparison itself, all four PanNuke-trained checkpoints were evaluated with no fine-tuning on the CPM-17 test split (32 H&E tiles, substantially larger and more variable in size than PanNuke's 256×256 patches). Because

CPM-17 provides no nuclear-class taxonomy comparable to PanNuke's five categories, evaluation here is binary (nucleus-versus-background), matching the standard protocol for this kind of cross-dataset test.

Configuration	Prior	Kept px (%)	GT recall (%)	CPM-17 Foreground Dice	Fwd. time (ms/img)
Dense U-Net	**none**	**100.0**	**100.0**	**0.836**	**94.3**
Sparse U-Net	raw (no mask)	100.0	100.0	0.813	396.0
Sparse U-Net	tissue (OD)	85.8	99.4	0.801	378.6
Sparse U-Net	hematoxylin	78.1	99.0	0.823	361.8

Table 2. CPM-17 external test results (32 tiles), using checkpoints trained exclusively on PanNuke fold 1 with no fine-tuning on CPM-17.

The ordering seen internally reproduces externally: the dense baseline is again both the most accurate (83.6% foreground Dice) and far faster (94.3 ms/image versus 362–396 ms/image for the sparse variants, roughly a 4× slowdown) than every sparse configuration. Forward times on CPM-17 are substantially higher than on PanNuke for all four configurations — consistent with CPM-17's larger tile size requiring more sliding windows per image, not a sparse-specific effect, since the dense model's time grows by a comparable factor. The relative ranking among the three sparse variants shifts slightly between the internal and external sets (hematoxylin edges out raw and tissue on CPM-17's foreground Dice, where it had trailed on PanNuke's nuclei-macro Dice); with only 32 external images this reordering is within the expected noise of a small test set and should not be over-interpreted. What does not change, and what this cross-dataset check was designed to test, is the central result: the dense baseline's advantage over every sparse configuration holds on data the model never saw during training or calibration.

6. Discussion

The central negative result — that patch-level stain-based sparsification underperforms a matched dense baseline on both speed and accuracy — is informative rather than merely disappointing, and is reported here deliberately rather than adjusted away. The likely explanation is a structural mismatch between the two domains the pipeline was moved across:

- **Occupancy.** In a CT volume windowed to bone-range Hounsfield units, bone occupies a small minority of voxels, so a sparse representation skips a large amount of genuinely empty computation. In a PanNuke patch, by contrast, tissue and nuclei together occupy the great majority of the field of view (the calibrated priors here retained 64–69% of pixels at 98% nuclear recall); there is comparatively little truly "empty" space at the patch scale to skip.
- **Overhead.** Sparse convolution requires indexing and bookkeeping (active-site lookup, rule generation for submanifold and strided sparse convolutions) that dense convolution avoids entirely by relying on a GPU's native ability to parallelize a full, regular 2D grid. When the retained fraction is not small, this bookkeeping cost is not repaid by a proportional reduction in useful computation, which is consistent with the 5–6× slowdown measured here relative to the dense baseline. The forward-pass times in Table 1 also exclude the one-time prior-mask computation (under 8 ms/image in every configuration); including it only widens the

sparse pipeline's end-to-end disadvantage, since this cost is paid in addition to, not instead of, the slower forward pass.

- **Where sparsity would likely help.** The natural analogue of "background" for a whole-slide image is not stroma or connective tissue within a diagnostic field, but the genuinely empty glass and out-of-focus regions surrounding tissue on a slide, which can occupy a large majority of a WSI's raw pixel area. A sparsity prior applied at whole-slide scale to separate tissue from background — rather than at patch scale to separate nuclei from stroma, as tested here — is the setting structurally closest to the CT bone case and is the natural next experiment.
- **Hardware and dimensionality are not matched to the source study.** The original CT result (a roughly 25× speedup) was measured for a 3D sparse U-Net on a full-size workstation GPU (RTX A6000); this study's 5–6× slowdown is for a 2D sparse U-Net on a smaller GPU (T4). The two speedup figures are not directly comparable magnitudes — they come from different network dimensionality, hardware, and problem scale — and the comparison that *is* controlled here is the internal one, between the sparse and dense variants trained and run under identical conditions.

7. Limitations and Future Work

1. **Patch-scale, not slide-scale.** This study operates on pre-extracted 256×256 PanNuke patches. The efficiency argument for sparsity is expected to strengthen substantially at true whole-slide-image scale, where background occupies far more of the raw pixel budget than it does within a curated patch.
2. **Semantic, not instance, segmentation.** The pipeline predicts per-pixel class labels and does not separate touching or overlapping nuclei into distinct instances, which is typically required for downstream cell-counting and morphometry; a distance-map or watershed post-processing stage would be needed for instance-level output.
3. **External validation is small and on a substitute dataset.** The cross-dataset check (Section 5.3) uses CPM-17 rather than the originally intended CoNSeP, substituted after CoNSeP's public download was placed behind an institutional sign-on with no available mirror; the two datasets are similar in kind (H&E nuclear instance segmentation, equivalent image-plus-mask format) but are not identical, and CPM-17's 32-tile test split is small enough that the exact ranking among the three sparse variants should be treated as noisy, even though the dense-versus-sparse headline result reproduces clearly. A larger external panel, and CoNSeP itself if institutional access becomes available, would strengthen this claim further.
4. **Single training budget.** All models were trained for a fixed, moderate budget (40 epochs × 150 batches) on one GPU. Absolute Dice values for all configurations, including the dense baseline, would likely improve with a larger training budget or model width; the comparative ordering between sparse and dense variants is the more load-bearing result here than any single absolute number.
5. **Class imbalance.** The "dead" nuclear class was not learned by any configuration and would need targeted class-balanced sampling or class-weighted loss terms to be modeled reliably.

8. Conclusion

This report performed a controlled, architecture-matched test of whether a sparsity-prior design that is effective for CT bone segmentation transfers to nuclei segmentation in digital pathology. Under identical training protocols, loss functions, and network capacity, a dense U-Net baseline was comparable to or ahead of every stain-prior-driven sparse variant on segmentation accuracy — the largest accuracy gap (dense vs. hematoxylin-prior sparse) is 0.030 nuclei-macro Dice, while dense vs. unmasked sparse differs by only 0.005, within the run-to-run noise this study can resolve — and decisively faster on inference speed in every case. The same ordering reproduced with no fine-tuning on an external dataset (CPM-17), indicating the speed result in particular is not an artifact specific to PanNuke. The report attributes this to the substantially higher pixel occupancy of stained tissue relative to bone in a CT volume, and identifies whole-slide-scale tissue/background separation as the setting in which the same architectural idea would be expected to actually pay off, forming a concrete direction for follow-up work. All code, calibration logic, training configuration, and evaluation metrics reported here are reproducible from the accompanying script.

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Prepared as a supporting technical report; full implementation available on request. Corresponding author: Dr. K. Mithra, Scottsdale, Arizona.